Aaradhya Dev Tamrakar

Statement of Purpose · Computer and Information Science Internship Programme

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1. Research Vision & Intellectual Objectives

Modern computational systems operate at the critical intersection of empirical machine learning, hardware-constrained execution, and formal safety guarantees. While deep neural architectures have achieved unprecedented empirical utility, their deployment in safety-critical, edge, or distributed environments is routinely undermined by three fundamental vulnerabilities: unquantified demographic failure modes, latent behavioral drift, and an absence of formal invariant bounds. My overarching research objective is to bridge empirical artificial intelligence with formal verification methods—designing learning systems that are quantifiably fair, resource-bounded, and mathematically verifiable. The Computer and Information Science internship at the Max Planck Society represents the preeminent environment to pursue this inquiry under the guidance of world-leading researchers in algorithms, systems, and machine learning.

2. Foundational Research & Applied Systems Experience

As a final-year undergraduate in Electronics, Communication and Information Engineering at Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University, my research and implementation work spans three interconnected pillars:

• Formally Verified Systems & SMT Invariant Proving: To address the unreliability of heuristic software bounds, I developed the Invariant Assurance Engine using the Microsoft Z3 SMT solver. The framework mechanically proves inductive safety properties across distributed state machines, verifying non-negative credit balances, monotonic lease transitions, and deadlock-freedom. In controlled adversarial experiments (INV-BMK-001), the engine achieved 100% planted violation recall (4/4 seeded failure modes detected and reproduced in sandbox execution) with zero false discoveries on valid specifications.

• Statistical Fairness Auditing & Mechanistic Evaluation: As a Fusemachines AI Fellow, I engineered BiasAperture, an algorithmic auditing harness evaluating demographic disparities across 126 intersectional demographic bins on large-scale visual corpora (97,000+ images). I implemented calibrated non-parametric Bootstrap confidence intervals (N=1,000, BCA method) and equalized odds disparity metrics, demonstrating how statistical audits expose localized degradation that global aggregate accuracy metrics systematically conceal.

• Deterministic Edge AI & Systems Programming: In SPARK (undergraduate capstone), I engineered FreeRTOS firmware on the Xtensa dual-core ESP32-S3 microcontroller, implementing ring-buffered I2C sensor acquisition from a 6-axis IMU. I designed and quantized an on-device neural classifier to an 18.5 KB INT8 footprint using TFLite Micro with strict sub-100ms deterministic latency bounds, achieving 0.9185 AUC-ROC on SisFall benchmark data with host-level Unity/CTest integration.

3. Alignment with Max Planck CIS Research Groups

The Max Planck Institutes provide unparalleled intellectual depth across the exact frontiers where my background lies. I am particularly drawn to research directions across MPI for Informatics (MPI-INF) and MPI for Software Systems (MPI-SWS):

• Reliable, Interpretable & Fair Machine Learning: Collaborating with groups investigating neurosymbolic learning, causal fairness, and certified robustness against distribution shifts, applying my empirical auditing experience from BiasAperture.

• Rigorous Systems Verification: Exploring the intersection of formal methods (SMT solving, model checking) and automated reasoning about autonomous or multi-agent pipelines, expanding upon my work with Z3-based invariant assurance.

• High-Performance & Resource-Constrained Inference: Investigating quantization, memory-bounded computation, and runtime verification on heterogeneous or embedded architectures.

4. Objectives for the Internship & Academic Trajectory

During the Summer 2027 internship, I aim to immerse myself in an active research group, formulate rigorous theoretical or empirical hypotheses, and build reproducible, open-source experimental artifacts suitable for peer-reviewed workshop or conference dissemination. Following the completion of my Bachelor of Engineering degree in January 2027, this internship

will serve as the foundational intellectual stepping stone toward pursuing my doctoral studies in computer science. I am eager to contribute my systems engineering stamina, empirical rigor, and mathematical enthusiasm to the Max Planck Society.